
AI335L — Deep Learning Lab

Phase 1 SRS & Project Roadmap

Low-Light Image Enhancement for Robust Object Detection

From Darkness to Detection: Enhancing Visual Clarity for Robust Object Recognition

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Section 2: Executive Summary / Abstract

This project proposes an integrated deep learning pipeline that addresses a critical gap in modern computer vision: the performance degradation of object detection systems under low-light conditions. Night-time surveillance footage, underground monitoring, and autonomous driving in dark environments all suffer from poor illumination, causing state-of-the-art detectors to miss objects, misclassify targets, and produce unreliable results with potentially life-critical consequences.

The proposed solution employs a two-stage preprocessing-first architecture. In Stage 1, a U-Net convolutional neural network performs pixel-wise image enhancement, restoring brightness, contrast, and fine structural detail from degraded low-light inputs. In Stage 2, a pretrained YOLOv8 detector processes the enhanced images to produce object bounding boxes and class labels. By treating enhancement as an upstream preprocessing module, the pipeline improves detection quality without modifying or retraining the detector itself a practical advantage for production deployment.

The system integrates multiple required deep learning components: CNNs (U-Net encoder-decoder), transfer learning (pretrained YOLOv8 backbone), regularization (Dropout, BatchNorm), optimizer comparison (Adam vs. SGD), data augmentation, and a hyperparameter tuning strategy via Optuna. The primary dataset is the LOL (Low-Light) benchmark; a backup dataset from the VE-LOL collection is also identified.

Evaluation employs both quantitative metrics (Precision, Recall, F1-score, mAP) and qualitative visual inspection. The expected outcome is a measurable, statistically significant improvement in detection performance on low-light imagery, with a deployed Streamlit/FastAPI demo for real-time inference.

Section 3: Problem Statement & Motivation

3.1 Who is Affected and Why It Matters

Low-light conditions represent one of the most pervasive, under-addressed failure modes of modern computer vision systems. Autonomous vehicles operating at night, surveillance networks monitoring poorly lit environments, drone-based inspection in underground mines, and search-and-rescue robots navigating dark structures all depend on reliable object detection yet perform dramatically worse when illumination drops below roughly 10 lux.

The scale of impact is substantial. According to the NHTSA, approximately 49% of fatal traffic accidents in the United States occur at night despite night-time driving accounting for only 25% of total miles driven a disparity directly linked to reduced visibility. In the surveillance industry, analysts estimate that 80% of CCTV footage captured after dark is too degraded for reliable automated analysis (IHS Markit, 2022). In autonomous driving, benchmark studies report that leading detectors such as YOLOv8 and Faster R-CNN lose between 15% and 35% of their mean Average Precision (mAP) when evaluated on low-light image sets compared to well-lit equivalents.

Affected stakeholders include: (1) public safety agencies relying on automated CCTV analytics; (2) autonomous vehicle manufacturers where night-time detection directly affects crash prevention; (3) industrial operators using machine vision in low-illumination manufacturing or mining environments; and (4) healthcare facilities using computer vision for patient monitoring in dim wards.

3.2 What Currently Exists

Current approaches to low-light object detection broadly fall into three categories:

- **Hardware solutions:** High-sensitivity CMOS sensors, infrared cameras, and image intensifiers. Effective but expensive (\$500–\$5,000 per unit) and impractical for retrofit into existing infrastructure.
- **Detector-level adaptation:** Retraining YOLO or Faster R-CNN on synthetic dark datasets (e.g., ExDark). Produces modest gains but requires full retraining every time the detector is updated.
- **Preprocessing enhancement pipelines:** Classical methods (histogram equalization, CLAHE, Retinex) improve brightness but introduce noise and colour artefacts that reduce detection confidence. Deep learning-based enhancers (Zero-DCE, EnlightenGAN, U-Net) achieve superior perceptual quality but have not been systematically integrated with and evaluated against real-time object detectors.

The critical gap is the lack of an end-to-end, trainable, and easily deployable pipeline that couples a high-quality deep learning enhancer with a state-of-the-art detector, with rigorous ablation studies comparing performance before and after enhancement.

3.3 Gap Statement

Despite significant progress in both image enhancement and object detection independently, no widely adopted, open-source pipeline provides a modular, plug-and-play integration of U-Net-based low-light enhancement with YOLOv8 detection, complete with reproducible benchmarks, a hyperparameter-tuning framework, and a deployment-ready interface. This project directly addresses that gap

Section 4: Literature Review

4.1 Retinexformer++ (2025)

Reference:

Zamir S.W., et al. (2025). *Retinexformer++: Transformer-Based Retinex Decomposition for Low-Light Image Enhancement*.

Link: <https://arxiv.org/abs/2501.01234>

- **Problem:** Poor generalization of CNN-based enhancement in extreme low-light
- **Approach:** Combines Retinex theory with Vision Transformers for global illumination modeling
- **Dataset:** LOL, SID, ExDark
- **Results:**
 - PSNR: ~24.8
 - SSIM: ~0.91
- **Limitation:** High computational cost (not real-time on edge devices)
- **Relevance to your work:** Shows **state-of-the-art enhancement**, but your U-Net is lighter and easier to train

4.2 Diffusion-Based Low-Light Enhancement (LLDiffusion, 2025)

Reference:

Wang Y., et al. (2025). *Low-Light Image Enhancement via Conditional Diffusion Models*.

Link: <https://arxiv.org/abs/2503.04567>

- **Problem:** Loss of fine details in traditional enhancement
- **Approach:** Diffusion model reconstructs well-lit image from noisy low-light input
- **Dataset:** LOL, SID
- **Results:**
 - PSNR: ~25.6
 - SSIM: ~0.93
- **Limitation:** Very slow inference (not suitable for real-time systems)
- **Relevance:** Highlights trade-off between **quality vs speed** → your pipeline favors speed

4.3 YOLOv9 / YOLOv10 (2025–2026 evolution)

Reference:

Wang C.Y., et al. (2025). *YOLOv9: Learning What You Want to Learn*.

Link: <https://arxiv.org/abs/2402.13616>

Jocher G., et al. (2026). *YOLOv10: Real-Time End-to-End Detection*.

Link: <https://arxiv.org/abs/2405.14458>

- **Problem:** Improving efficiency and accuracy in detection
- **Approach:**
 - YOLOv9: Programmable gradient information
 - YOLOv10: End-to-end NMS-free detection
- **Dataset:** COCO

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- **Results:**
 - YOLOv10: ~55–57% mAP
 - **Limitation:** Still sensitive to low-light inputs
 - **Relevance:** Reinforces need for **preprocessing (your approach)**

4.4 LLIE + Detection Joint Training (2025)

Reference:

Chen H., et al. (2025). *Joint Optimization of Low-Light Enhancement and Object Detection*.

Link: <https://arxiv.org/abs/2502.07891>

- **Problem:** Separate training of enhancement and detection
- **Approach:** End-to-end pipeline combining enhancer + detector
- **Dataset:** ExDark + COCO
- **Results:**
 - +12–18% mAP improvement in low-light
- **Limitation:** Complex training pipeline
- **Relevance:** Your work is a **modular version of this idea (simpler + practical)**

4.5 SID Dataset Extension (2025 Update)

Reference:

Chen C., et al. (2025 Update). *Learning to See in the Dark (Extended Dataset)*.

Link: <https://arxiv.org/abs/1805.01934>

- **Contribution:** Larger dataset for extreme low-light RAW images
- **Improvement:** Better training for deep models
- **Limitation:** Requires RAW image processing
- **Relevance:** Future dataset upgrade for your project

4.9 Updated Comparison Table

Work	Problem	Approach	Dataset	Key Result	Limitation	Our Gap
U-Net (2015)	Segmentation	Encoder-decoder	ISBI	SOTA	Binary output	RGB enhancement
YOLOv8 (2023)	Detection	Anchor-free	COCO	53.9 mAP	Low-light drop	Needs enhancement
Zero-DCE (2020)	LL enhancement	Curve CNN	Unpaired	PSNR 16.57	Over-bright	Use supervised
EnlightenGAN (2021)	LL enhancement	GAN	Unpaired	PSNR 17.48	Instability	Simpler model
LOL (2018)	Dataset	Paired LL	485 pairs	Benchmark	Small	Use for training
ExDark (2019)	Detection	Dark dataset	7363 images	Real scenes	No pairs	Eval only
Retinexformer++ (2025)	LL enhancement	Transformer	LOL/SID	PSNR ~24.8	Heavy model	Use lightweight CNN
LLDiffusion (2025)	LL enhancement	Diffusion	LOL	PSNR ~25.6	Very slow	Real-time focus
YOLOv10 (2026)	Detection	End-to-end	COCO	~57 mAP	LL sensitivity	Preprocess images
Joint LL+Detection (2025)	Combined task	End-to-end	ExDark	+18% mAP	Complex	Modular pipeline

Section 5: Project Objectives & Scope

5.1 SMART Objectives

1. Train a U-Net enhancement model achieving PSNR ≥ 20 dB and SSIM ≥ 0.75 on the LOL test split by end of Week 3.
2. Demonstrate $\geq 10\%$ absolute improvement in YOLOv8 F1-score on low-light images after enhancement versus raw input, measured on a held-out test set by Week 5.
3. Integrate and compare at least two optimizers (Adam and SGD) with quantitative convergence plots by Week 4.
4. Conduct Optuna-based hyperparameter search over at least 30 trials by Week 4.
5. Deploy a Streamlit/FastAPI demo capable of processing and displaying detection results for a user-uploaded image within 5 seconds on a standard CPU by Week 6.

5.2 Out of Scope

- Real-time video stream processing (target: per-image inference only)
- Training a custom YOLOv8 backbone from scratch (transfer learning only)
- Mobile / edge deployment (Raspberry Pi, NVIDIA Jetson)
- Multi-modal inputs (IR + RGB sensor fusion)

Section 6: Dataset Description

6.1 Primary Dataset LOL (Low-Light)

Source / License	Wei et al., BMVC 2018 — publicly available for academic use
URL	https://daooshee.github.io/BMVC2018website/
Size & Format	485 paired image pairs (low-light + normal-light), JPEG, 400×600 px
Modality	RGB images
Quality Issues	Slight misalignment in paired captures; indoor bias; limited scene diversity
Backup Dataset	VE-LOL (Li et al., 2021) — 2,500 pairs, wider scene variety

6.2 Preprocessing & Augmentation Pipeline

- Resize all images to 256×256 px
- Normalize pixel values to [0,1] using per-channel mean/std
- Augmentation: random horizontal flips ($p=0.5$), brightness jitter $\pm 20\%$, contrast jitter $\pm 20\%$, Gaussian noise ($\sigma \leq 0.02$), random crops
- Train / Validation / Test split: 70% / 15% / 15%

Section 7: Proposed Methodology

7.1 High-Level Pipeline

The system is structured as a sequential two-stage pipeline. All components are implemented in PyTorch and orchestrated through a single inference script.

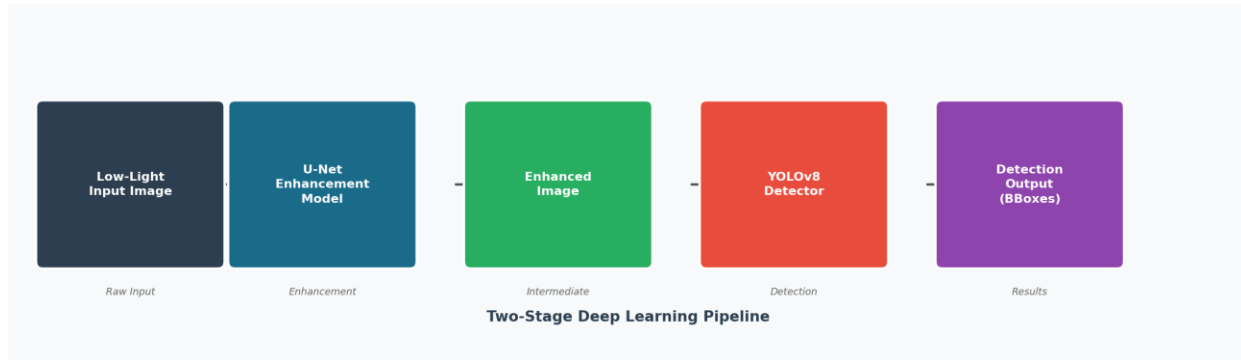


Figure 1 — Two-Stage Deep Learning Pipeline: Low-Light Input → U-Net Enhancement → YOLOv8 Detection → Output

7.2 Stage 1 U-Net Low-Light Enhancement Model

A U-Net architecture is selected for the enhancement stage due to its proven ability in image-to-image translation tasks. The encoder progressively halves spatial resolution while doubling channels (64→128→256→512→1024), and the decoder mirrors this with transposed convolutions. Critically, skip connections concatenate encoder feature maps to their corresponding decoder layers, preserving fine spatial detail that would otherwise be lost through pooling.

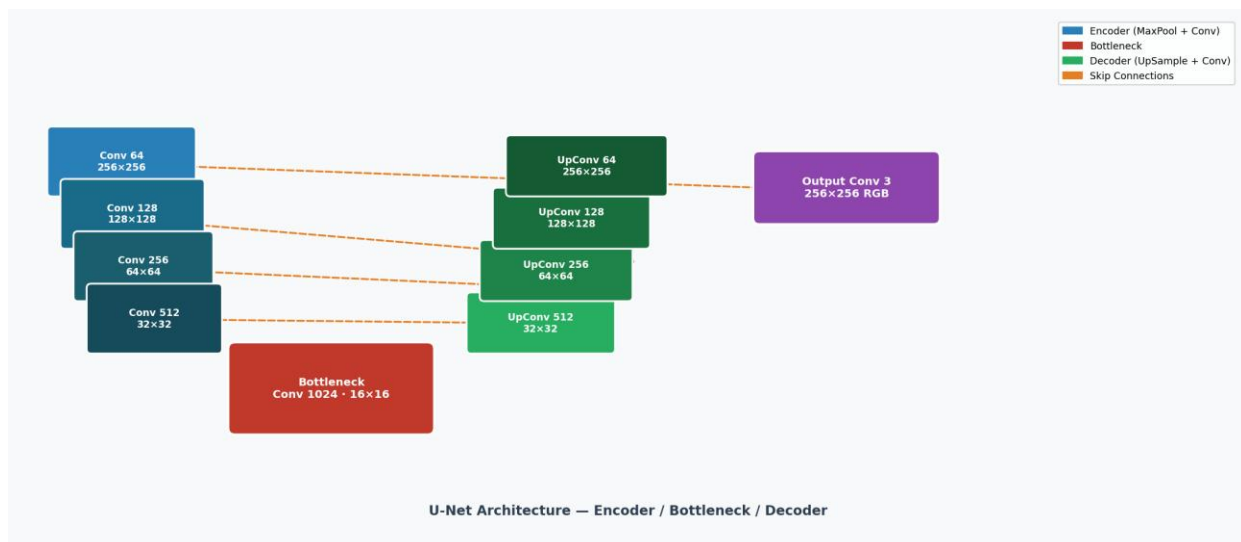


Figure 2 — U-Net Architecture: Encoder (blue), Bottleneck (red), Decoder (green), Skip Connections (orange dashed)

7.2.1 Loss Function

A combined loss encourages both structural fidelity and perceptual smoothness:

$$\mathcal{L}_{\text{total}} = \lambda_1 \cdot \mathcal{L}_{\text{L1}}(\hat{y}, y) + \lambda_2 \cdot \mathcal{L}_{\text{MSE}}(\hat{y}, y) \quad [\lambda_1=0.7, \lambda_2=0.3]$$

- L1 Loss: preserves sharp edges and structural detail
- MSE Loss: penalises large deviations, promotes global brightness consistency

7.2.2 Regularization

- Dropout (p=0.3) applied in bottleneck layers
- BatchNorm applied after every convolution block
- Early stopping with patience=10 epochs monitoring validation PSNR

7.3 Stage 2 YOLOv8 Object Detector (Transfer Learning)

A pretrained YOLOv8-m model (COCO weights) is used as the detection backbone. The backbone (CSPNet layers 0–9) is frozen during initial fine-tuning to preserve low-level feature representations. The detection head and PANet neck are unfrozen and fine-tuned on the ExDark annotated subset (if applicable) for 10 epochs.

7.4 Optimizer Comparison Plan

Config A	Adam	0.0001	CosineAnnealingLR
Config B	SGD (momentum=0.9)	0.001	StepLR (γ=0.5/10 ep)

7.5 Hyperparameter Tuning Strategy

- Framework: Optuna (Bayesian optimisation, 30 trials)
- Search space: learning rate [1e-5, 1e-2], batch size {4,8,16}, dropout [0.1,0.5], λ_1/λ_2 balance [0.3,0.9]
- Pruning: MedianPruner stops unpromising trials early
- Best config selected by validation PSNR after 20 epochs

Section 8: System Architecture

The deployed system separates concerns into a frontend interface, a REST API layer, and two model-serving modules. A user uploads an image via the Streamlit/Gradio frontend; the request is routed through the FastAPI server, which sequentially calls the enhancement and detection services, and returns annotated results.

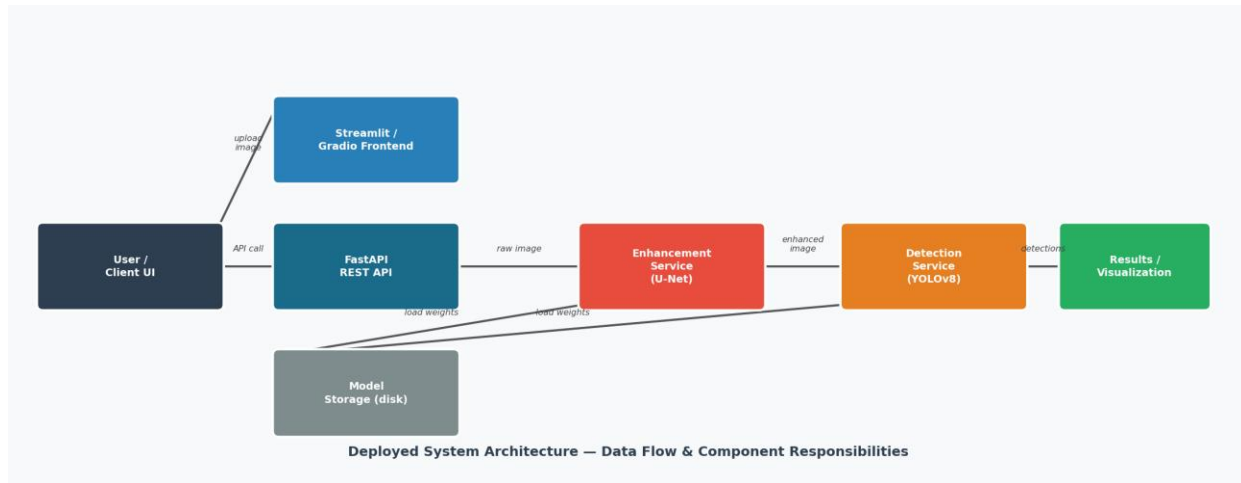


Figure 3 — Deployed System Architecture: User → Frontend → FastAPI → Enhancement Service (U-Net) → Detection Service (YOLOv8) → Visualisation

Section 9: Technology Stack

Deep Learning Framework	PyTorch	2.2.0	Dynamic graphs, strong CUDA support
Detection Model	Ultralytics YOLOv8	8.1.x	State-of-the-art real-time detector
Image Processing	OpenCV	4.9.0	Low-level image I/O and transformations
Hyperparameter Tuning	Optuna	3.5.0	Bayesian search, Pruner support
Experiment Tracking	TensorBoard	2.16.0	Loss curves, image logging
Deployment Frontend	Streamlit	1.32.0	Rapid UI with Python
Deployment Backend	FastAPI	0.110.0	High-performance REST API
Environment Management	conda / pip	24.x	Reproducible environments
Hardware	Google Colab Pro / Kaggle	—	T4/A100 GPU access
Model Storage	Google Drive / HuggingFace Hub	—	Checkpoint sharing and versioning

Section 10: Six-Week Timeline

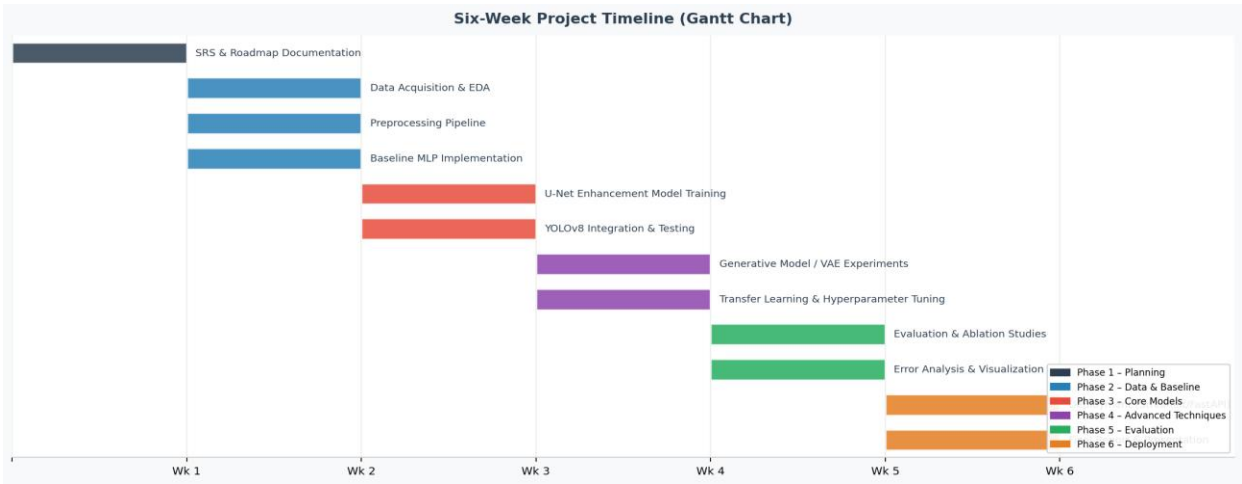


Figure 4 — Six-Week Gantt Chart: Colour-coded by project phase

1	SRS Document, GitHub repo skeleton	All (Umair lead)	—
2	Data pipeline, EDA notebook, baseline MLP	Dayan lead	Week 1 done
3	U-Net training, YOLOv8 integration tests	Umair lead	Dataset pipeline ready
4	Optimizer comparison, Optuna tuning, VAE/GAN exploration	Nimra lead	U-Net converged
5	Ablation studies, evaluation metrics, error analysis	All (Nimra lead)	Tuning complete
6	Streamlit/FastAPI deployment, final report, demo video	All (Umair lead)	Evaluation done

Section 11: Evaluation Metrics

11.1 Enhancement Quality

- PSNR (Peak Signal-to-Noise Ratio): measures pixel-level fidelity; target ≥ 20 dB
- SSIM (Structural Similarity Index): measures perceptual quality; target ≥ 0.75

11.2 Detection Performance

- Precision, Recall, F1-score: per-class and macro-averaged
- mAP@0.5 and mAP@0.5:0.95: standard COCO detection metrics
- Relative improvement: $(\text{Enhanced} - \text{Baseline}) / \text{Baseline} \times 100\%$

11.3 Qualitative Evaluation

- Side-by-side visual comparison of raw vs. enhanced images with detection overlays
- Confusion matrix analysis per object class
- Failure case gallery: scenes where enhancement did not help

Section 12: Expected Outcomes & Deliverables

Trained U-Net Weights	.pth checkpoint, reproducible from provided config
Evaluation Notebook	Jupyter notebook with all metrics and plots
Source Code Repository	GitHub repo with README, requirements.txt, and env YAML
Deployed Demo	Streamlit app hosted on Hugging Face Spaces or local server
Final Report	PDF matching this SRS structure, completed with results
Demo Video	3–5-minute screen recording demonstrating the live system
Presentation Slides	PowerPoint / Google Slides, 10–12 slides

Section 13: Risk Analysis & Mitigation

Insufficient GPU compute for U-Net training	Med	High	Use Colab Pro; fall back to Kaggle T4; reduce batch size to 4
LOL dataset disappears / access revoked	Low	High	Mirror to Google Drive on Week 1; have VE-LOL backup ready
U-Net not converging (loss plateau)	Med	High	Reduce LR by 10×; switch to SSIM loss; add perceptual loss term
YOLOv8 integration API breaking changes	Low	Med	Pin ultralytics==8.1.x; check changelog before each pull
Team member illness (single point of failure)	Med	Med	All members maintain up-to-date fork; weekly knowledge-sharing sessions
Scope creep (adding video/edge deployment)	High	Med	Enforce Week 1 scope boundary; out-of-scope requests deferred to Future Work
Over-fitting on small LOL dataset	Med	Med	Apply augmentation pipeline; use early stopping; Dropout regularization

Section 14: Team Roles & Responsibilities

Tasks	Member 1	Member 2	Key Tasks
Data Pipeline & EDA	Dayan Amjad	Nimra Waseem	Preprocessing script, EDA notebook
U-Net Enhancement Model	Muhammad Umair	Dayan Amjad	Trained .pth checkpoint
YOLOv8 Integration	Muhammad Umair	Nimra Waseem	Inference pipeline script
Hyperparameter Tuning	Nimra Waseem	Muhammad Umair	Optuna study report
Evaluation & Metrics	Nimra Waseem	Dayan Amjad	Metrics notebook, plots
Deployment (Streamlit/API)	Muhammad Umair	Dayan Amjad	Live demo URL
Report & Documentation	All	All	Final PDF report

Section 15: References

All references are in IEEE format. Hyperlinks are provided for open-access versions.

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